# Grounded coupler design | EC 165 / 250 / 160 / 260 MHz

Revised after mandatory dressed-idle calibration - 29 October 2026. Hard OFF limit remains below 10 GHz. All frequencies are cyclic frequencies (H/h), capacitances are fF. The earlier large integrated chi mismatch was an uncompensated-control diagnostic; it is superseded as the final readout result.

| Item | Result |
|---|---|
| Three-mode exchange OFF | 9.980174 GHz; modest-basis residual convergence: True |
| Five-mode exchange OFF | 9.800131 GHz before resonator integration |
| Calibrated seven-mode OFF | 9.902329 GHz; physical 0100-0010 null at reported basis. 1 Hz certification remains unresolved. |
| Earlier seven-mode CZ | 119.930 ns; raw 99.7952%; postselected 99.9986%. PRIOR controls; not recalibrated after idle retuning. |
| Calibrated integrated readout | Both pairs pass nominal constraints and complete independent +/-1% EJ sign grids. A mismatch 0.0767%, B mismatch 0.6975%. |

## Confirmed input constraints

| Input | Value |
|---|---|
| Loaded rail EC/h | A1 165; A2 250; B1 160; B2 260 MHz - immutable at every circuit stage |
| Idle operating-frequency guides | A 4.61 GHz; B 5.39 GHz. Individual bare settings compensate dressing; final dressed means are reported separately. |
| Isolated-pair logical splitting | A 150 MHz; B 90 MHz at the design reference; fit once and freeze hardware |
| Gate scope | About 100 ns static CZ; A stationary, B moved; B20 physical auxiliary (0020). Minimum-S1 toy selection owns logical source. |
| Coupler bounds | Loaded EC 100-600 MHz; ON >7 GHz; ON < OFF <10 GHz |
| Loading | Per coupler: each rail <10%; prefer <7%, minimize combined loading. Coupler own self-capacitance >=55% of effective budget. Four-branch aggregate reconstructed separately. |
| Topology | Floating rails; C2 directly grounded, C1 active. Actual A22-C1, C1-B11, A22-B11 capacitors. Equal pad-ground caps within each rail transmon; self caps compensate loaded EC. No unspecified parasitics. |
| Readout | Independent A/B designs first; nominal 50 ohm; logical-state mean readout targets 5.990/6.010 GHz. EJ/EC >45 exception. Mandatory individual-EJ idle calibration after integration. |
| Numerical targets | Residual couplings/ZZ: 1 Hz absolute; gate errors and survival: 1e-4 absolute probability. Finite audits stop with explicit nonconvergence. |

# 1 | Toy selection and fresh three-body search

| Quantity | Value |
|---|---|
| Selected ON channel | Logical 10 to B20 |
| Stationary A / nominal moved B | 4.610000 / 4.830473 GHz |
| Common intended dressed coupling | 5 MHz; nominal resonant cycle 100 ns |
| Toy S1 / S2 | 0.00343080 / 0.00573366 |
| Maximum unintended \|g/detuning\| | 0.060798 (<0.2) |
| Toy calibrated time / raw / postselected | 102.942 ns / 99.3951% / 99.9917% |

Logical 0/1 are the lower/upper isolated-pair one-excitation modes. Pair vacuum is separate. The physical B20 label does not determine the logical source. All four logical sources were compared at common 5 MHz using the established full-cosine toy score basis.

## Three-body search

Fresh independent Ca/Cb ratios from 0.125 to 8, three initial EC values and interior OFF seeds were used. The signed exchange was solved for the actual direct capacitor, giving 42 verified seed families; 11 independent continuous optimizations then minimized rail loading while retaining the OFF branch and ON-strength seed.

| Parameter | Three-mode solution |
|---|---|
| A22-C1 / C1-B11 / direct A22-B11 | 10.151256 / 12.738629 / 0.946167 fF |
| Loaded coupler EC | 414.346891 MHz |
| ON / verified retuned OFF | 7.000100 / 9.980174 GHz |
| Reduced 11-to-02 ON strength seed | 8.836696 MHz (not a logical gate certificate) |
| Rail-to-C loading A2 / B1 | 2.4143% / 1.8339% |
| Coupler own fraction | 55.0010% |

OFF nulls physical A2/B1 01-10, with C in its ground reference and virtual coupler paths included. The larger reduced ON matrix element compensates pair-hybridization charge overlaps when targeting 5 MHz in the selected logical channel. The search minimum is near the allowed upper OFF boundary; it is a local best found design, not a global proof.

# 2 | Five-body refinement and capacitances

A1/B2 are restored without resonators. Positive self capacitances and intra-pair links reproduce the fixed loaded ECs and A150/B90 idle splittings. The first 9.98015 GHz five-body design shifted outside 10 GHz after readout integration; it was rejected at validation. Returning to five-body design with a 9.8 GHz search ceiling supplied integration margin without relaxing any user constraint.

| Actual physical capacitor | Five-mode fF | Integrated seven-mode fF |
|---|---|---|
| A22 - C1 | 6.564568 | 6.564568 |
| C1 - B11 | 14.372491 | 14.372491 |
| A22 - B11 | 0.786207 | 0.786207 |
| A12 - A21 | 13.668593 | 21.849062 |
| B12 - B21 | 6.517997 | 6.609151 |

| Self capacitor | Five-mode fF | Seven-mode fF |
|---|---|---|
| A1 across pads (C1-ground for C) | 55.832396 | 101.258654 |
| A2 across pads (C1-ground for C) | 34.071822 | 4.218413 |
| C across pads (C1-ground for C) | 23.660171 | 25.549591 |
| B1 across pads (C1-ground for C) | 56.409294 | 103.593851 |
| B2 across pads (C1-ground for C) | 35.747691 | 8.682536 |

| Pad-ground capacitor (each pad) | Five-mode fF | Seven-mode fF |
|---|---|---|
| A1 | 117.395329 | 23.479066 |
| A2 | 77.480917 | 131.717559 |
| B1 | 121.063933 | 24.212787 |
| B2 | 74.500882 | 126.651499 |

Loaded coupler EC = 450.284990 MHz. Maximum compensated relative EC error in five-mode design: 1.68e-16. Coupler capacitors are unchanged during readout integration; one-time self-cap and intra-pair compensation is then frozen. C2 is exactly ground, not a large capacitor approximation.

# 3 | Loading and OFF residuals

| Budget quantity | A2 | B1 | Coupler |
|---|---|---|---|
| Five-mode complete-branch loading | 2.2761% | 2.1739% | 44.9990% |
| Five-mode own / total effective C | 93.9745% | 96.5946% | 55.0010% |
| Seven-mode own / total effective C | 90.4445% | 95.5695% | 59.3932% |
| Seven-mode rail-to-C loading | 2.0459% | 2.3848% | 40.6068% |

Own capacitance is across-pad self C plus the series combination of the two pad-ground capacitors (C1-ground self C for the grounded coupler). Ceff = e^2/(2h EC). Loading is 1 - EC_loaded/EC_without_links, holding self caps fixed. The complete coupler branch includes the direct rail capacitor; the rail-to-C-only counterfactual removes the two C links. These are different reported budgets, not interchangeable percentages.

| Shared rail | Four physical branches | C-link loading | Own fraction | EC MHz |
|---|---|---|---|---|
| A2 | fixed hardware | 6.6663% | 88.7128% | 236.0024 |
| A2 | shared pair EC compensated | 7.0617% | 88.0434% | 250.0000 |
| B1 | fixed hardware | 6.7129% | 91.4879% | 151.5412 |
| B1 | shared pair EC compensated | 7.0876% | 91.0128% | 160.0000 |

Four-branch results reconstruct the nodal circuit with four physical coupler/opposite-pair branches attached to the same central rail. They are not four times the single-branch percentage. The compensated row is a separate static capacitance budget, not a four-neighbor dynamic gate simulation.

| Modes | OFF GHz | Physical g Hz | 01L-10L Hz | 00L-11L Hz | ZZ Hz |
|---|---|---|---|---|---|
| Five | 9.800131 | 7.88e-07 | 832.24 | 820.37 | -111.08 |
| Seven | 9.902329 | 8.52e-07 | 852.18 | 840.34 | 7.50 |

Coupler EJ/h at OFF: five modes 29.295865 GHz; seven modes 29.880200 GHz. These are the bare coupler f01 coordinates, not dressed normal-mode eigenfrequencies.

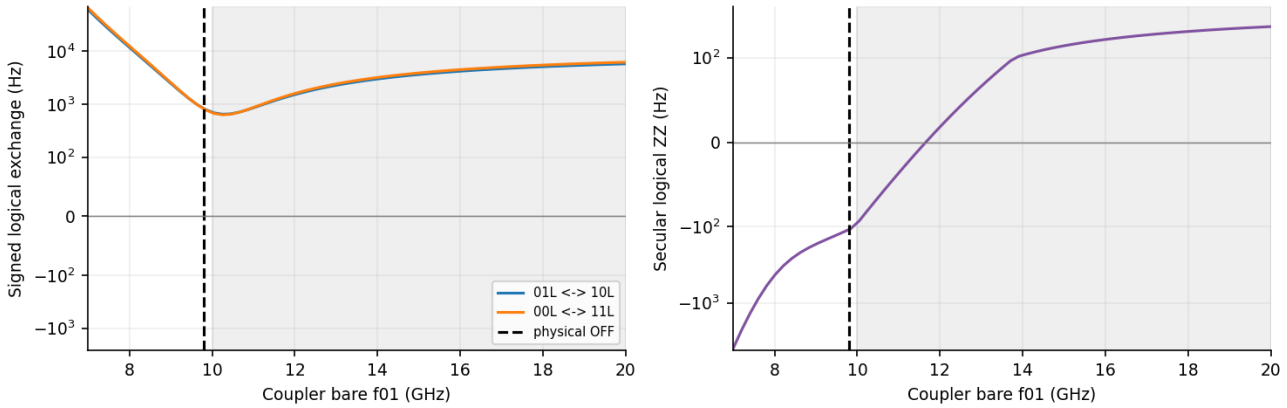
Connected physical-state detuning at OFF: five modes -755.5376 MHz; seven modes -783.5500 MHz. Logical 01L-10L detuning in seven modes is -52.5468 MHz.

Five/seven-mode OFF temporarily nulls physical 0100-0010 after canonical projection onto all four physical single-excitation references. Logical exchanges and secular ZZ remain residual diagnostics. This temporary physical-versus-logical choice remains an open workflow question. Seven-mode values above are basis-dependent estimates, not 1 Hz-certified residuals.

# 4 | Exchange detunings and diagnostic sweep

| Connected states at seven-mode idle OFF | g (kHz) | Detuning (MHz) | \|g / detuning\| |
|---|---|---|---|
| 01 <-> B20 | -239.6825 | -624.0762 | 0.000384 |
| 00 <-> B20 | -231.7416 | -717.5767 | 0.000323 |
| 11 <-> B20 | 139.1831 | -478.0290 | 0.000291 |
| 00 <-> B02 | -113.5820 | -587.9270 | 0.000193 |
| 10 <-> B20 | 96.1147 | -571.5295 | 0.000168 |
| 10 <-> B02 | 75.5464 | -441.8797 | 0.000171 |
| 01 <-> B02 | 67.2709 | -494.4264 | 0.000136 |
| 11 <-> B02 | -47.7774 | -348.3792 | 0.000137 |

Small exchange magnitudes should be assessed alongside detunings. Detunings are diagonal differences in the reported canonical/local-energy frame. Secular ZZ uses tracked logical eigenbranches: (E11-E10-E01+E00)/h.




Five-mode sweep uses local5/pair12/C5, ncut30 and frozen capacitors/idle controls; it illustrates shape rather than Hz-level pointwise convergence. Grey shading above 10 GHz is diagnostic only. The dashed line marks the retuned five-mode physical null. Signed logarithmic axes show both signs and small residuals.

# 5 | Prior static CZ benchmarks - unchanged controls

These gate results are retained from BEFORE the new integrated idle calibration. They are not a recalibrated gate using the newly adjusted stationary A rail. The revised idle cell records those new stationary settings for future ON calibration; no new gate fidelity is claimed in this revision.

| Quantity | Five modes | Seven modes |
|---|---|---|
| ON coupler f01 (GHz) | 7.050000 | 7.168900 |
| Moved B bare f01 (GHz) | 4.871278 | 4.862037 |
| Duration (ns) | 110.162 | 119.930 |
| Conditional-phase error (rad) | 8.78e-12 | 2.13e-09 |
| Raw CZ fidelity | 99.5145% | 99.7952% |
| Postselected score | 99.9990% | 99.9986% |
| Mean logical survival | 99.5155% | 99.7967% |
| Worst-input nonlogical population | 0.7902% | 0.3934% |
| Actual-bias max unintended ratio | 0.065893 | 0.089909 |
| Fixed-control gate precision | <1e-4 between tested bases | True |

| Logical input | Five-mode nonlogical population | Seven-mode nonlogical population |
|---|---|---|
| 00 | 0.4686% | 0.1843% |
| 01 | 0.7902% | 0.3934% |
| 10 | 0.3103% | 0.1884% |
| 11 | 0.3688% | 0.0472% |

The conditional phase is calibrated to pi before improving return. Geometry, loaded ECs and charge coefficients remain frozen; coupler EJ, moved B-pair EJ/frequency and duration may vary. A remains fixed. Starts lie within 200 MHz of the preceding solution. The finite duration search is 70-130 ns, around the requested 100 ns target; this is a local numerical search, not a global optimum proof.

All scores use consistent static ON dressed references, retaining CZ auxiliaries. There is no sudden idle-to-ON jump or ramp in this benchmark. Raw and postselected values refer to the SAME operating point. Postselected score = raw score / average logical survival, matching the toy convention; it is a survival-normalized benchmark, not a separately optimized gate. No decoherence, readout drive or pulse-shape error is included.

# Optional EJ-pulse CZ | definition and scope

Added 2026-11-03; optional post-design study, requested separately. Earlier static-gate and integrated-readout sections retain their original controls, dates and validation status. This section reads completed calculations; no optimization was run during reporting.

**What is calculated**

Coherent evolution of the nominal FIVE-transmon A1-A2-C-B1-B2 circuit. Start at the saved five-mode OFF position (C f01 = 9.800131 GHz), with A at 4.61 GHz and B at 5.39 GHz. Freeze all capacitances, loaded ECs and charge couplings; A EJs remain fixed. Move C, B1 and B2 simultaneously to gate settings, hold, then return symmetrically. The C ON endpoint is fixed at 7.05 GHz. Calibrate the common bare B gate frequency and hold duration at conditional phase pi, retaining the selected logical 10 to B20 CZ channel.

Loaded EC/h = (165, 250, 450.285, 160, 260) MHz in physical order A1,A2,C,B1,B2. Use the SAME four full-system dressed IDLE logical states as both inputs and final projections. The full rise, hold, fall and simulated ringing act on them. This differs from static ON-state preparation. No resonators, T1, flux noise or measurement dynamics are included.

**Rise time means 10% to 90% of the commanded EJ excursion**

For normalized rising envelope s(t), the quoted rise is t(s=0.9)-t(s=0.1). It is NOT the full edge duration T. Falling edges use the same width. Both shapes interpolate EJ directly; local frequencies follow the full-cosine spectrum at fixed loaded EC. These are not tanh/cosine frequency pulses or calibrated SQUID-flux waveforms.

$$s_{\tanh}(t) = \frac{\tanh[(t - T/2)/\tau] + \tanh 4}{2\tanh 4}, \quad T = 8\tau$$

$$t_{10-90} = 2\tau\,\mathrm{atanh}(0.8\tanh 4), \quad T = 3.645897\,t_{10-90}$$

$$s_{\cos}(t) = \frac{1 - \cos(\pi t/T)}{2}, \quad T = 1.693955\,t_{10-90} \quad (0 \le t \le T)$$

| Commanded rise | Tanh full edge T | Cosine full edge T |
|---|---|---|
| 10 ns | 36.46 ns | 16.94 ns |
| 1 ns | 3.65 ns | 1.69 ns |

**1 GHz sharp cutoff and timing convention**

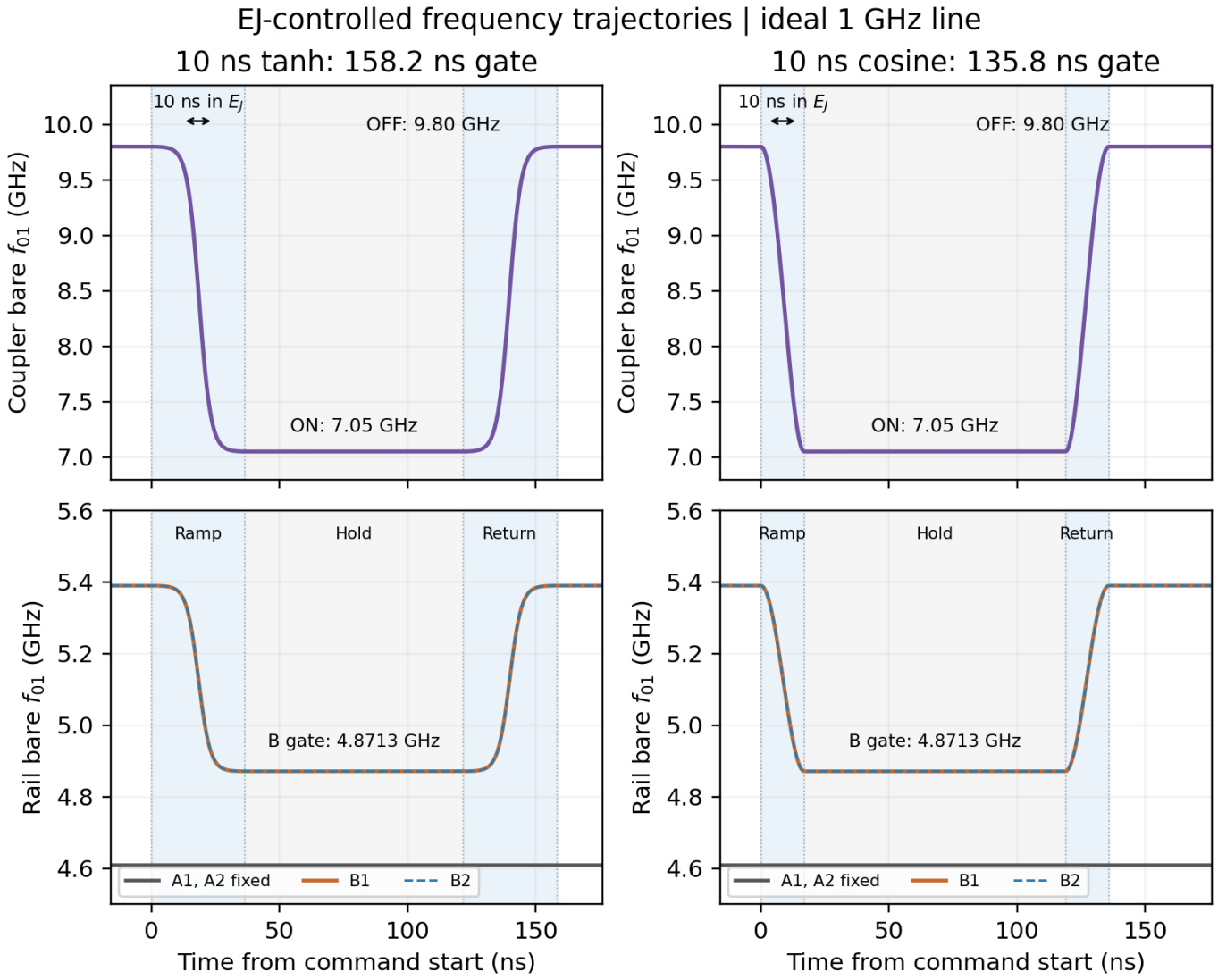
Filter the ENTIRE EJ envelope with H(f)=1 for |f|<1 GHz and zero above it, assuming zero phase (a pure delay is removed). Retain overshoot and pre/post-ringing without clipping. The line model acts on EJ; a physical flux line requires a separate nonlinear SQUID map.

$$E_{J,i}(t) = E_{J,i}^{\mathrm{idle}} + \left(E_{J,i}^{\mathrm{gate}} - E_{J,i}^{\mathrm{idle}}\right) s_{\mathrm{filtered}}(t)$$

Programmed gate duration = 2T + hold. Propagation initially adds 16 ns before and after that span, and the fastest-edge audit tests 32 ns per side. This padding is a numerical window, not a mandatory hardware guard. An ideal brick wall has infinite sinc tails and no exact finite settling time. The programmed and simulated durations are both retained in the data.

# Optional EJ-pulse CZ | frequency tuning in time

Representative saved 10 ns tanh and cosine gates. Colored intervals mark the complete EJ ramp, calibrated hold and return; gray lines mark command boundaries. The 10 ns arrows refer to 10%-90% of EJ excursion, not to the full shaded ramp.




These are BARE local transmon frequencies obtained from each filtered EJ waveform and its loaded EC. They are not dressed logical splittings or instantaneous coupled-system eigenbranches. A1/A2 remain fixed at 4.61 GHz; B1/B2 go from 5.39 GHz to approximately 4.8713 GHz; C goes from its five-mode OFF setting at 9.8001 GHz to 7.05 GHz. The two B curves nearly coincide; their underlying EJ excursions differ.

| 10 ns command | Full ramp (ns) | Hold (ns) | Programmed total (ns) |
|---|---|---|---|
| tanh | 36.46 | 85.28 | 158.20 |
| cosine | 16.94 | 101.93 | 135.81 |

The 1 GHz filter and pre/post-ringing are included in the curves. The shown traces use saved calibrated endpoints; no gate optimization or coupled time propagation was rerun.

# Optional EJ-pulse CZ | calibration and results

Each row is one locally calibrated phase-pi pulse. The bounded search maximizes RAW fidelity; the postselected score and survival are evaluated at that SAME pulse. Common B gate f01 is about 4.8713 GHz, with the precise per-row values saved. This is not a global optimum proof.

| Shape | Rise (ns) | Hold (ns) | Programmed total (ns) | Raw (%) | Selected (%) |
|---|---|---|---|---|---|
| tanh | 10 | 85.28 | 158.20 | 99.551 | 99.991 |
| tanh | 5 | 97.57 | 134.03 | 99.489 | 99.878 |
| tanh | 2 | 102.02 | 116.60 | 99.151 | 99.662 |
| tanh | 1 | 105.18 | 112.47 | 99.047 | 99.598 |
| tanh | 0.5 | 106.79 | 110.44 | 99.000 | 99.579 |
| cosine | 10 | 101.93 | 135.81 | 99.493 | 99.986 |
| cosine | 5 | 107.49 | 124.43 | 99.432 | 99.856 |
| cosine | 2 | 105.80 | 112.58 | 99.096 | 99.644 |
| cosine | 1 | 107.09 | 110.47 | 99.031 | 99.593 |
| cosine | 0.5 | 107.70 | 109.39 | 98.989 | 99.578 |

**Fidelity and postselection convention**

Let M be the four-by-four final propagator projected onto the dressed idle logical basis. After the same permitted local-Z phase corrections used throughout the CZ workflow, raw fidelity includes nonlogical loss. The selected value is the established survival-normalized score, not a separately optimized gate.

$$\bar{p} = \frac{\mathrm{Tr}(M^\dagger M)}{4}, \quad F_{\mathrm{raw}} = \frac{\mathrm{Tr}(M^\dagger M) + |\mathrm{Tr}(U_{\mathrm{CZ}}^\dagger Z M)|^2}{20}, \quad F_{\mathrm{sel}} = \frac{F_{\mathrm{raw}}}{\bar{p}}$$

The 10 ns cosine gate takes 135.81 ns versus 158.20 ns for 10 ns tanh: 22.39 ns shorter. Their postselected scores differ by about 0.0052 percentage point, below the adopted 0.01-point probability-accuracy target; the raw-score difference is about 0.0574 point. At similar total duration, 5 ns tanh has visibly lower selected fidelity than 10 ns cosine.

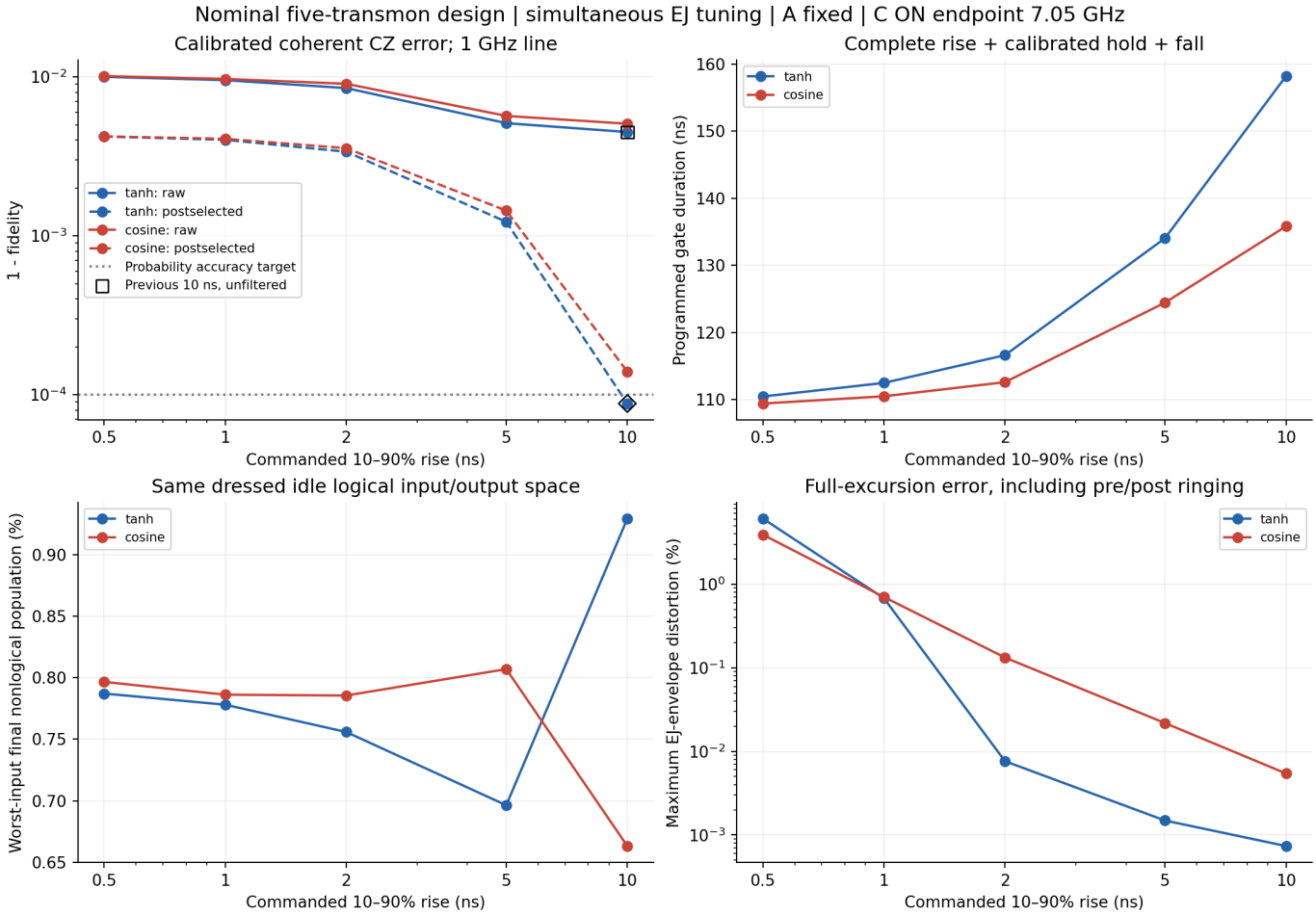
Shorter edges increase coherent wrong-logical-state population as well as leakage. Postselection removes nonlogical population but cannot remove logical mixing. From 1 to 0.5 ns the programmed gate shortens by only 1-2 ns. The delivered 0.5 ns edges broaden to about 0.59 ns (tanh) and 0.57 ns (cosine). The 1 GHz filter barely changes the 10 ns tanh result.

**Numerical scope**

Start with six levels per rail transmon, five coupler levels and a total local-index cutoff of six (287 even-parity states). Transfer phase at fixed hold/C to eight rail levels, seven coupler levels and cutoff eight (782 states). Cutoff is a numerical truncation, not excitation-number conservation. Propagation keeps full charge interactions and off-diagonal EJ-dependent cosine terms in one fixed basis. Filter FFT: 4096 ns span, 1/256 ns sample. The final 0.5 ns rows use a 0.0375 ns integration step; other rows use approximately 0.075 ns.

# Optional EJ-pulse CZ | comparison figure

Same five-transmon hardware and dressed idle frame; ideal 1 GHz EJ cutoff. Original four-panel layout, scaled to fit this portrait page.



# Optional EJ-pulse CZ | checks and timing

**Recorded costs from the first comparison**

| Shape / rise | Main case (s) | Filtered trials | Screened pulse propagation (s) |
|---|---|---|---|
| tanh / 10 ns | 324.7 | 4 | 48.4 |
| tanh / 5 ns | 262.2 | 4 | 42.6 |
| tanh / 2 ns | 215.1 | 4 | 36.3 |
| tanh / 1 ns | 189.5 | 4 | 36.5 |
| tanh / 0.5 ns | not fully timed | 30 | 275.6 |
| cosine / 10 ns | 183.4 | 4 | 41.4 |
| cosine / 5 ns | 218.6 | 4 | 39.3 |
| cosine / 2 ns | 201.5 | 4 | 37.7 |
| cosine / 1 ns | 191.7 | 4 | 34.4 |
| cosine / 0.5 ns | 291.4 | 17 | 148.9 |

Main case includes unfiltered seeding, filtered calibration and larger-basis phase transfer, but excludes later focused checks. The two shape workers ran concurrently; summed durations are not wall time. Exact end-to-end wall time was not recorded. The last tanh search was stopped after 30 trials; its logged trial propagation totals 275.6 s and fixed-hold completion 139.0 s. The focused cosine audit took 197.2 s; its final fine-step phase correction took 43.6 s.

**Why it was slow**

The ideal line response adds ringing throughout the pulse, including the plateau. Changing hold duration changes the filtered waveform, so the filtered calculation cannot simply reuse a constant ON spectrum. Every trial propagated four logical inputs through half the complete time-symmetric waveform. A screened trial took roughly 8-13 s; a logged larger-basis propagation typically took 26-49 s. Larger-basis transfer repeated such propagations to enforce phase pi. The long 0.5 ns tanh line search kept sampling a fidelity plateau despite a five-iteration limit.

**Representative numerical checks**

Representative 0.5 ns cosine checks: step halving changed raw fidelity by 0.01457 percentage point and selected fidelity by 0.000157 point; the finer row was adopted and rephased. Larger excitation cutoff changed raw by 0.000078 point; doubled tail window changed raw by 0.000429 point. These latter checks used the earlier coarse controls/step. Some worst-input leakage differences exceeded $10^{-4}$. A full independent $10^{-4}$ certificate for every row, or a further step audit at each final fine control, was not completed. The 135 regression tests passed.

# Optional EJ-pulse CZ | speedup options

**Already implemented after this run**

Hard 12-trial filtered search limit; fidelity/survival/leakage plateau stop at $10^{-4}$ after eight trials; bounded phase finish; cached phase-slope/secant updates instead of a fresh derivative probe each iteration. Two shape workers, one BLAS thread each, exact waveform caches and time symmetry were already used. Default integration step is now 0.0375 ns for rises<=1 ns and 0.075 ns otherwise. The exact as-run driver and interrupted-case completion are preserved. The revised routine has not been rebenchmarked across the full sweep.

**Further proposals - not implemented in this report**

1. Reuse the closest calibrated edge as the next seed, instead of repeating a seven-frequency scan plus scalar polish for every case.
2. Test the filtered pulse at the unfiltered optimum first. If differences are below $10^{-4}$, try phase-only transfer before a new two-control search.
3. Reuse identical fixed-basis Hamiltonian matrices and idle states across rise times; benchmark the smallest adequate production basis on representative slow/fast edges.
4. Use a separately checked adaptive propagation step to concentrate work near edges/ringing. Do not replace a ringing plateau by a constant Hamiltonian without validating that approximation.

These proposals need separate agreement and measurement. No new basis reduction or pulse approximation was applied for this PDF, and no speed factor is promised. Reproduction inputs, case records, comparison.csv/json, figure and provenance are in output/pulse-shape-comparison-1GHz/.

# 6 | Independent and integrated readout

Each standalone design contains only its own rail pair plus one resonator. Each uses two equal actual inner-pad-to-resonator links. The isolated-pair target is A150/B90; dressed splitting acceptance is A140-160/B80-100 MHz. No coupler or opposite rail was present during link fitting.

| Readout component | A pair | B pair |
|---|---|---|
| Each actual rail-resonator link (fF) | 9.337256 | 4.527794 |
| Bare resonator LC f (GHz) | 6.081051 | 6.056678 |
| Resonator self C (fF) | 523.445508 | 525.551904 |
| Resonator L (nH) | 1.308614 | 1.313880 |
| Standalone mean logical readout target | 5.990 GHz | 6.010 GHz |
| Standalone all workflow constraints | [] | [] |

| Constraint | Required |
|---|---|
| Chi mismatch | \|chi+ - chi-\| / \|(chi+ + chi-)/2\| <1% |
| Readout strength | Absolute signed mean chi >=1 MHz |
| Dispersivity / H / overlap | Worst \|g/detuning\| <0.1; H >=0.95; reference overlap >=0.95 |
| Transmon limits | Loaded EC 150-400 MHz; EJ/h <25 GHz; EJ/EC >45 |
| Frequency / stability | Dressed operating mean below dressed resonator; chi signs stable on independent +/-1% EJ1/EJ2, 5x5 grid |
| Physical implementation | Equal readout links; exact nodal inverse; fixed loaded EC through self-cap compensation |

## Coupler-resonator dispersivity at the prior seven-mode ON bias

| Resonator | Coupler transition | g (MHz) | Detuning (MHz) | \|g / detuning\| |
|---|---|---|---|---|
| r1 | 0-1 | 0.3213 | 1181.923 | 0.0002719 |
| r1 | 1-2 | 0.4368 | 644.333 | 0.0006779 |
| r1 | 2-3 | 0.5073 | 60.226 | 0.008424 |
| r2 | 0-1 | 16.8197 | 1160.888 | 0.01449 |
| r2 | 1-2 | 22.8642 | 623.298 | 0.03668 |
| r2 | 2-3 | 26.5547 | 39.191 | 0.6776 |

These local charge-matrix estimates use the prior benchmark ON bias. They are distinct from the full multilevel worst dispersive ratio used to validate readout.

# 6b | Calibrated integrated idle readout

Exchange OFF does not remove unequal self-energy shifts of A2 and B1. Individual rail EJs are now calibrated with all geometry and loaded ECs fixed; the common coupler OFF root is then rechecked. Both rails are evaluated jointly at the resulting settings. Near resonance is assessed by the existing readout H-score and chi constraints, not by imposing equal bare frequencies.

| Transmon | Original bare GHz | Calibrated bare GHz | Calibrated EJ/h GHz |
|---|---|---|---|
| A1 | 4.613060962 | 4.612856563 | 17.339380 |
| A2 | 4.613323398 | 4.624675459 | 11.953997 |
| B1 | 5.392300206 | 5.427281304 | 24.432179 |
| B2 | 5.391125368 | 5.390259268 | 15.423145 |

Common OFF: 9.853409358 -> 9.902328515 GHz. Capacitors, resonator L, loaded ECs and all charge coefficients remain unchanged. Only rail EJs and coupler OFF EJ change.

The unequal bare settings compensate unequal interaction-induced shifts. They are not the final pair eigenfrequencies: each rail still has distinct lower/upper logical transitions, with the splitting reported below. Equal bare frequencies do not establish dressed resonance, and exchange cancellation does not remove local frequency dressing.

| Comparison | A chi mismatch | B chi mismatch |
|---|---|---|
| Standalone readout | 0.990% | 0.990% |
| Integrated, uncompensated diagnostic | 11.132% | 27.709% |
| Integrated, calibrated final centers | 0.0767% | 0.6975% |

The large uncompensated mismatch therefore does not establish an intrinsic inability to read out the coupled circuit. The required individual-EJ calibration restores both nominal readout constraints and sign stability. This calibration is now mandatory in the workflow; historical retuned results must not be compared with unretuned new controls as a hardware comparison.

# 6c | Readout at calibrated idle centers

## Rail A at the saved nominal idle center

| Metric | Standalone | Calibrated integrated |
|---|---|---|
| Mean dressed operation (GHz) | 4.61 | 4.61001 |
| Logical splitting min (MHz) | 145.867 | 146.047 |
| Worst dispersive ratio | 0.0695464 | 0.0707847 |
| Chi mismatch (%) | 0.99 | 0.0767174 |
| H score minimum | 0.979989 | 0.982667 |
| Minimum \|mean chi\| (MHz) | 1.005 | 1.0298 |
| Minimum overlap | 0.987685 | 0.981372 |
| Constraint failures | none | none |

## Rail B at the saved nominal idle center

| Metric | Standalone | Calibrated integrated |
|---|---|---|
| Mean dressed operation (GHz) | 5.39 | 5.38999 |
| Logical splitting min (MHz) | 87.2733 | 93.5005 |
| Worst dispersive ratio | 0.0789341 | 0.0891257 |
| Chi mismatch (%) | 0.99 | 0.697539 |
| H score minimum | 0.971009 | 0.950206 |
| Minimum \|mean chi\| (MHz) | 1.005 | 1.12784 |
| Minimum overlap | 0.983322 | 0.976143 |
| Constraint failures | none | none |

Both calibrated centers pass the complete independent +/-1% EJ1/EJ2, 5x5 sign grids, including reference-overlap reliability, in the recorded adaptive probe bases. Integrated values are worst over both spectator logical states. Sign grids are not repeated at each frequency-map point; nominal trace acceptance away from the centers is not a fresh sign certificate.

# 6d | Standalone three-state readout contrast

Added 2026-11-03. Standard readout characterization after the independent readout links are fitted. Existing design, integrated-readout and gate sections retain their original controls, dates and validation status; this addition does not rerun them.

**Models, axes and state definitions**

Separate A1-r1-A2 and B1-r2-B2 systems; no coupler and no opposite rail. All capacitors and resonator inductances are fixed at the saved nominal designs. Each pair's two EJs vary independently on a 61 x 61 grid, with axes converted to bare single-transmon f01 using the loaded EC and full-cosine spectrum. Ranges are A: 4-5 GHz and B: 4.4-5.8 GHz. Stars mark the standalone nominal idle points.

00 denotes pair ground; 10 and 01 denote the lower and upper isolated-pair one-excitation branches used as overlap references. They are not unmixed physical occupations. The cavity frequency in state s is (E[s,1]-E[s,0])/h. All plotted contrasts are absolute differences of these frequencies, with no factor of two.

**Nominal standalone contrasts (screening basis)**

| Pair | 10 - 00 (MHz) | 01 - 00 (MHz) | \|01 - 10\| (kHz) |
|---|---|---|---|
| A | -0.998188 | -0.994630 | 3.558 |
| B | -0.999742 | -1.005117 | 5.375 |

**Contours and interpretation**

The following six-panel figure shows |01-10|, |01-00| and |10-00| at 0.5, 1 and 2 MHz. Orange means ALL three contrasts exceed 0.5 MHz; pink means ALL exceed 1 MHz. Blue marks the logical contrast below 10 kHz in the left column. Gray flags minimum reference assignment overlap below 95%; raw values remain saved. These are spectral-contrast diagnostics, not additional readout acceptance constraints or predictions of measurement fidelity. The existing H-score, chi-mismatch and dispersive constraints still apply separately.

**Basis checks and reproduction**

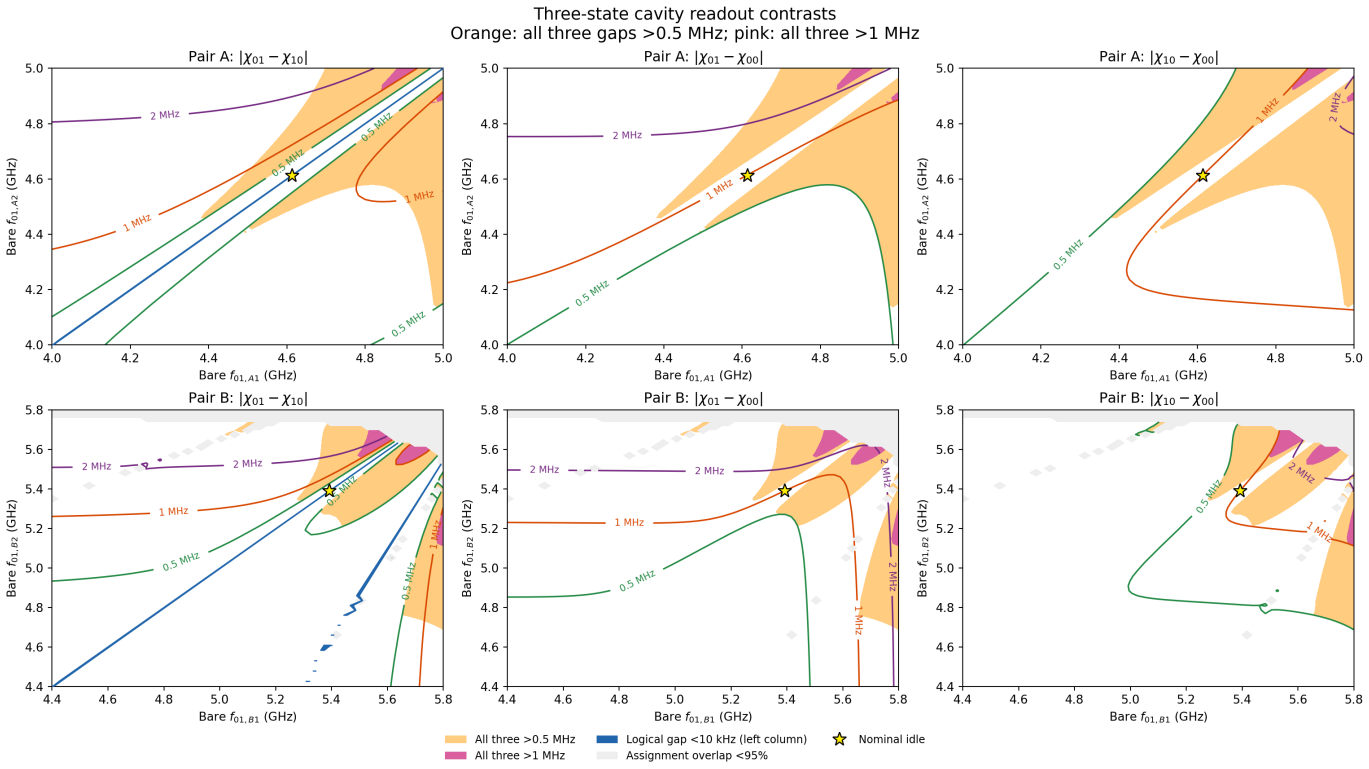
| Pair | Grid + audits (s) | Largest audited cavity-shift change (kHz) |
|---|---|---|
| A | 17.60 | 0.430 |
| B | 17.80 | 0.065 |

Screening uses six local transmon levels, resonator states 0/1/2, charge cutoff 25 and 30 assigned eigenlevels. Ten fixed-control checks per pair (nine grid locations plus nominal idle) use eight local levels, charge cutoff 35 and 40 eigenlevels, with the same three resonator states. Finite-grid contours and these spot checks do not certify every narrow boundary or Hz-level residual coupling. Nominal screening values above need not equal earlier larger-basis acceptance tables.

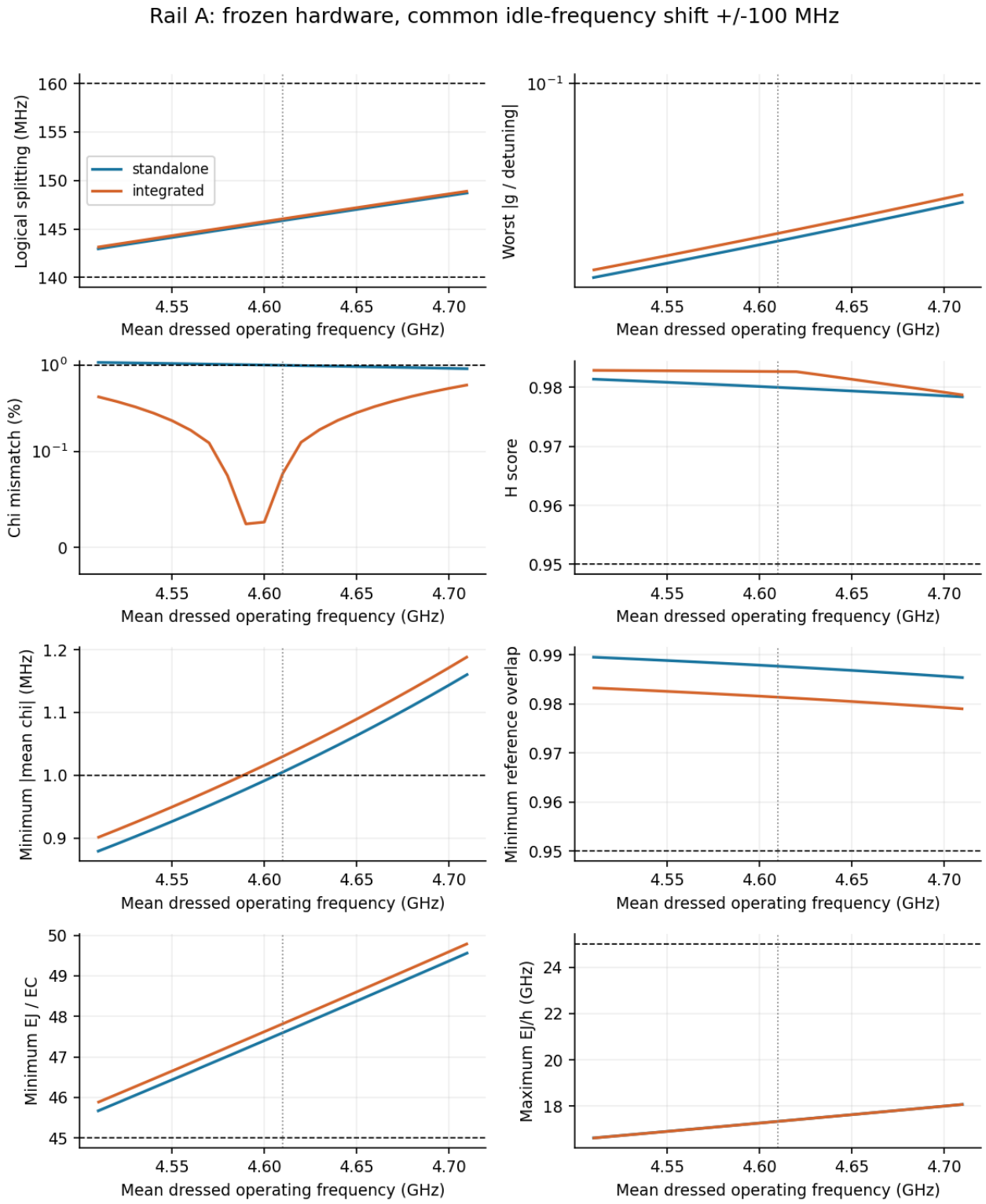
Source: readout-cavity-EJ-maps/{A,B}-summary.json and {A,B}-maps.npz, from nominal-design/fabrication-input.json readout hardware. Effective modal capacitors in the map snapshots are reductions of the physical floating-pad circuit, not fabricated link values. Replot with readout_shift_map.py; no optimization is run during report assembly. Seven-mode 2D maps are optional and excluded here.

# 6d | Standalone three-state readout contrast | maps

Original six-panel layout | Standalone triples | Bare f01 axes

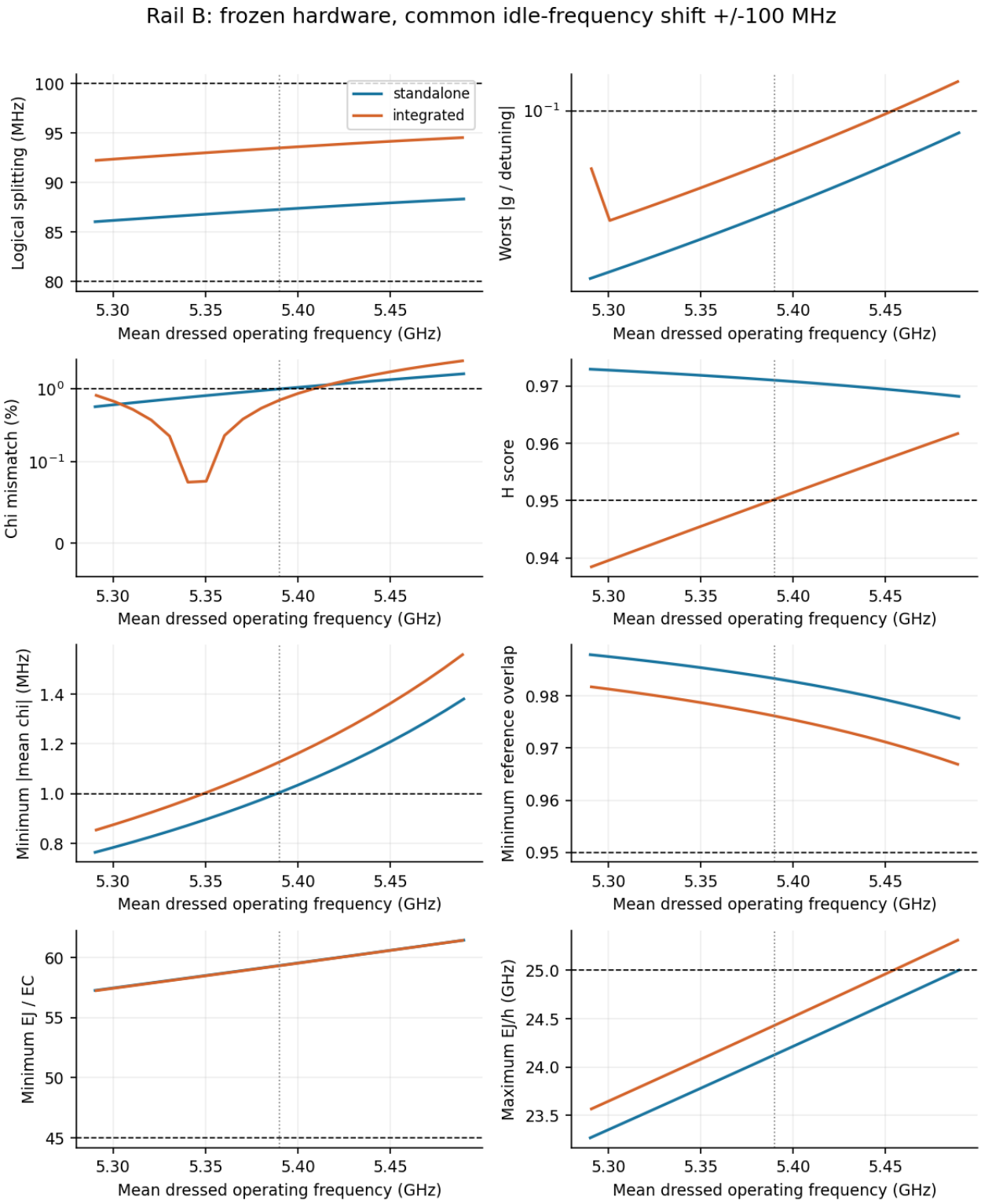


# 7A | Readout constraints over +/-100 MHz




Each trace varies both selected bare transmon frequencies together over +/-100 MHz around its own calibrated center. Their differential setting, all capacitors, resonator L, opposite rail EJs and coupler OFF EJ are fixed. The x coordinate is the resulting mean dressed operating frequency for each trace. Black dashed lines are actual thresholds; the grey line marks the intended operating-frequency guide. There is no new narrow frequency bound and no per-point retuning.

# 7B | Readout constraints over +/-100 MHz




Each trace varies both selected bare transmon frequencies together over +/-100 MHz around its own calibrated center. Their differential setting, all capacitors, resonator L, opposite rail EJs and coupler OFF EJ are fixed. The x coordinate is the resulting mean dressed operating frequency for each trace. Black dashed lines are actual thresholds; the grey line marks the intended operating-frequency guide. There is no new narrow frequency bound and no per-point retuning.

# 8 | Numerical checks, timing and reproducibility

| OFF check at identical controls | Physical exchange change | Logical exchange change | ZZ change |
|---|---|---|---|
| Five, last two tested bases | 0.199 Hz | 0.121 Hz | 0.024 Hz |
| Seven, search to modest validation | 277.567 Hz | 135.108 Hz | 32.806 Hz |

Five-mode OFF checks stop when physical exchange, both logical exchanges and secular ZZ agree within 1 Hz. The seven-mode audit table above belongs to the earlier unretuned reference; it did not meet 1 Hz. Current OFF residuals are freshly evaluated at calibrated controls with local6/pair12/C5 and three photon levels, but no new 1 Hz convergence certificate is claimed. Calibrated nominal readout and sign-stability checks pass. Full isolation and a recalibrated gate under the revised idle controls remain separate validation tasks.

Gate fixed-control checks compare raw fidelity, postselected score, mean survival and per-input leakage. Seven-mode maximum tested changes: {'raw_cz_fidelity': 4.212416759963311e-06, 'postselected_cz_fidelity': 1.0231657732173716e-07, 'logical_survival': 4.110366472409588e-06, 'worst_input_leakage_change': 1.3755454171704429e-05}. Photon levels remain three; a separate photon-cutoff convergence certificate is not supplied.

# 9 | Stage timings and saved records

| Stage / retained attempt | Measured elapsed seconds |
|---|---|
| toy selection and calibration seconds | 2.094 |
| three body search seconds | 12.460 |
| five body design seconds | 2.407 |
| five body off validation and sweep seconds | 10.774 |
| five body gate calibration seconds | 28.953 |
| seven body integration and off seconds | 46.370 |
| Three-body fixed-control audit | 2.099 |
| Independent readout A | 7.011 |
| Standalone/integrated map A | 268.075 |
| Independent readout B | 13.755 |
| Standalone/integrated map B | 268.671 |
| Prior seven-mode gate recalibration | 543.149 |
| Dressed-idle calibration, final refinement | 63.972 |
| Calibrated idle sign-grid A | 37.131 |
| Calibrated readout map A | 225.571 |
| Calibrated idle sign-grid B | 36.893 |
| Calibrated readout map B | 225.714 |
| Rejected near-10GHz five-mode attempt (design + audit + gate) | 52.444 |

Stage durations use perf_counter, including numerical search and verification in that stage. The interrupted/restarted seven-mode gate stage instead uses OS process-creation UTC to final endpoint time, plus its measured extra audit. Readout maps and the final gate job ran concurrently, so their durations must not be summed as total elapsed wall time. Preparation, code inspection, plotting/layout and the earlier superseded 12 GHz run are outside these numerical stage timings. The first failed seven-mode root scan took about 6.1 s process wall time. An earlier unseeded exploratory gate trial was discarded and is outside the reported final calibration timer; it is not a gate result.

Updated records: idle-retuned/final/result.json, calibrated-cell.json, post-retune-off.json, sign-A/B.json and map-A/B.json. Original cell.json, seven-off.json and map-A/B.json remain uncompensated diagnostics; seven-gate.json remains the prior-control benchmark. Hardware reconstruction and loaded ECs are unchanged. Final idle-calibration timing excludes discarded preparatory attempts; all shown map/sign jobs are timed independently and may overlap. Workflow and channel policy remain the authorities.